23.2 Radioactive Decay

Question Paper

Course	CIE A Level Physics
Section	23. Nuclear Physics
Topic	23.2 Radioactive Decay
Difficulty	Medium

Time allowed: 50

Score: /40

Percentage: /100

Question 1a

A student sets up a radioactive source and a Geiger counter to investigate how the count rate of the source varies with time.

State and explain how the student could demonstrate

(i)
the spontaneous nature of radioactive decay,

[2]

(i)
the random nature of radioactive decay.

[2]

[4 marks]

Question 1b

Technetium-101 $\left({}^{101}_{43}\mathrm{Tc}\right)$ decays by beta-minus emission to form a stable isotope of Ruthenium (Ru).

Complete the equation for this decay.

$${}^{101}_{43}\mathrm{Tc} \rightarrow {}^{\cdots\cdots}_{\cdots\cdots}\mathrm{Ru} + {}^{\cdots\cdots}_{\cdots\cdots}\beta$$

[2 marks]

Question 1c

The variation with time *t* of the number *N* of technetium-101 nuclei in a sample of radioactive material is shown in Fig. 1.1.

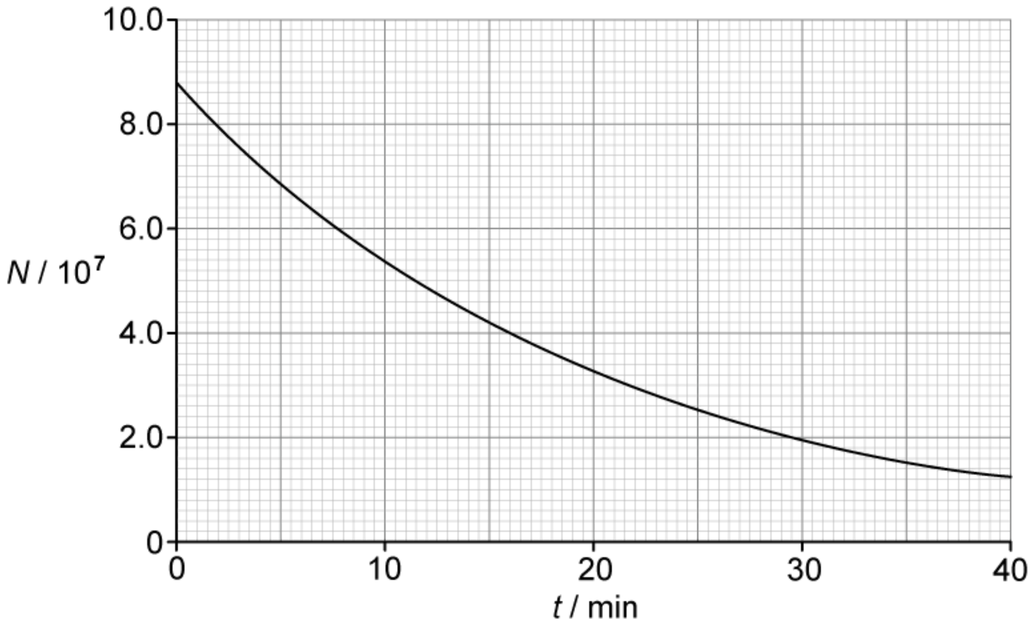


Fig. 1.1

(i)

Use Fig. 1.1 to determine the activity, in Bq, of the sample of technetium-101 at time $t = 14.0$ minutes.

[4]

(ii)

Use your answer in **(c)(i)** to calculate the decay constant λ of technetium-101.

[2]

(iii)

On Fig. 1.1, sketch a line to show the variation with *t* of the number of ruthenium nuclei in the sample.

[2]

[8 marks]

Question 1d

Each decay releases a beta particle with energy 487 keV.

(i)
Calculate, in J, the total amount of energy given to beta particles that are emitted between time t = 10 min and time t = 30 min.
[3]

(ii)
Suggest why the total amount of energy released by the decay process between time t = 10 min and time t = 30 min is actually greater than your answer in **(d)(i)**. [1]

[4 marks]

Question 2a

A source of water is found to be contaminated with the radioactive isotope radium-228 $\left({}^{228}_{88}\mathrm{Ra}\right)$.

This isotope of radium has a half-life of 5.75 years.

Explain the meaning of half-life in this context.

[2 marks]

Question 2b

Calculate the decay constant, in s^{-1}, of radium-228.

[2 marks]

Question 2c

A sample is taken of the contaminated water. The activity of radium-228 in a sample of 1.0 kg of water is found to be 20 mBq.

The mass of water in 1.0 mol is 18 g.

Calculate

(i)
the number of radium-228 atoms in 1.0 kg of the contaminated water

[2]

(ii)
the ratio

$$\frac{\text{number of molecules of water in 1.0 kg of water}}{\text{number of } {}^{228}\text{Ra atoms in 1.0 kg of water}}$$

[2]

[4 marks]

Question 2d

The maximum safe limit for the activity of radium-228 in water has been set as 18.5 mBq kg^{-1}.

Calculate the time, in days, for the activity of the contaminated water to be reduced to the safe limit.

[3 marks]

Question 3a

State what is meant by radioactive decay.

[2 marks]

Question 3b

A radioactive sample consists of an isotope with a half-life T that decays to form a stable product. Only the isotope and the stable product are present in the sample.

At time $t = 0$, the sample has an activity of A_0 and contains N_0 nuclei of the isotope.

(i)
On Fig. 1.1, sketch the variation with t of the number N of nuclei of the isotope present in the sample from time $t = 0$ to time $t = 3T$.

[3]

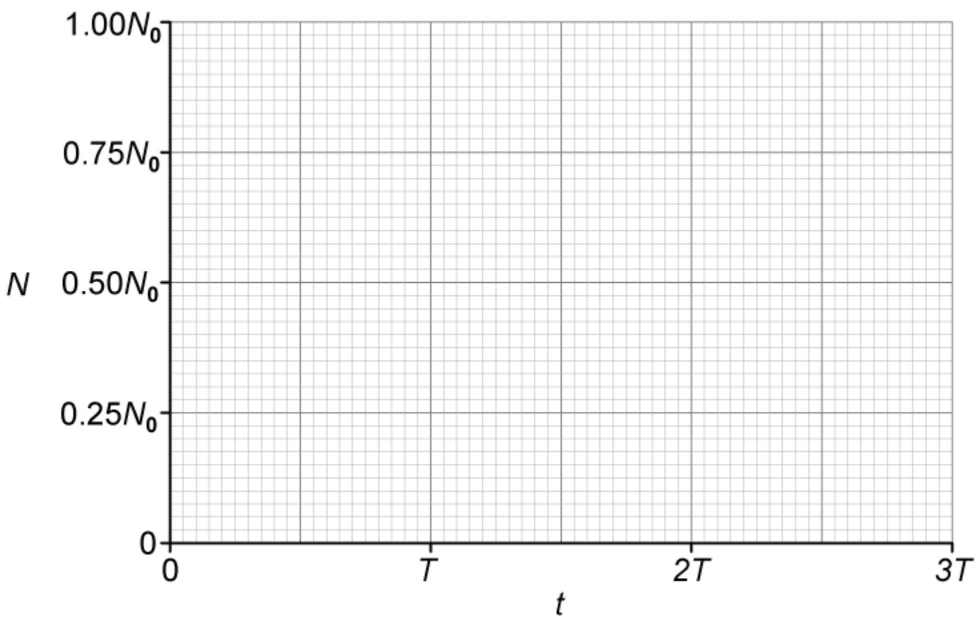


Fig. 1.1

(ii)
State the name of the quantity represented by the gradient of the graph in Fig. 1.1.

[1]

[4 marks]

Question 3c

(i)

On Fig. 1.2, sketch the variation with N of the activity A of the sample for values of N between N = 0 and $N = N_0$.

[2]

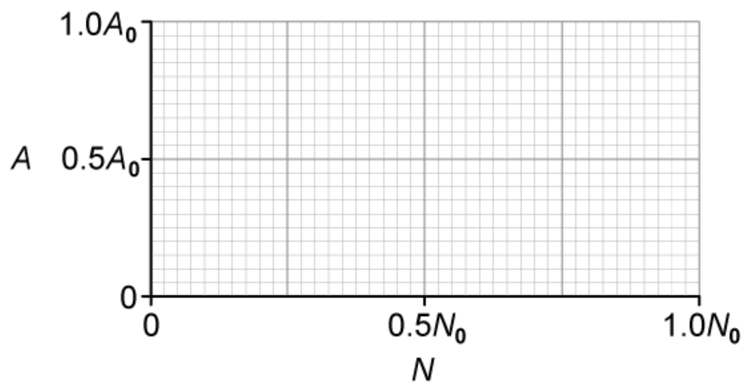


Fig. 1.2

(ii)

State the name of the quantity represented by the gradient in Fig. 1.2.

[1]

[3 marks]

Question 3d

Calculate the fraction of undecayed nuclei N remaining relative to N_0 at time $t = 1.25T$.

[2 marks]

Model Answers: Medium

1a

(a)

(i) The student could demonstrate the spontaneous nature of radioactive decay by:

- Repeating the experiment in different (environmental) conditions e.g. temperature, pressure, humidity, chemicals; [1 mark]
- To show that the rate of decay / activity does not change; [1 mark]

OR max **one mark** for:

- (Spontaneous decay means) the rate of decay / activity (of a nucleus) is not affected by environmental / external factors; [1 mark]

(ii) The student could demonstrate the random nature of radioactive decay by:

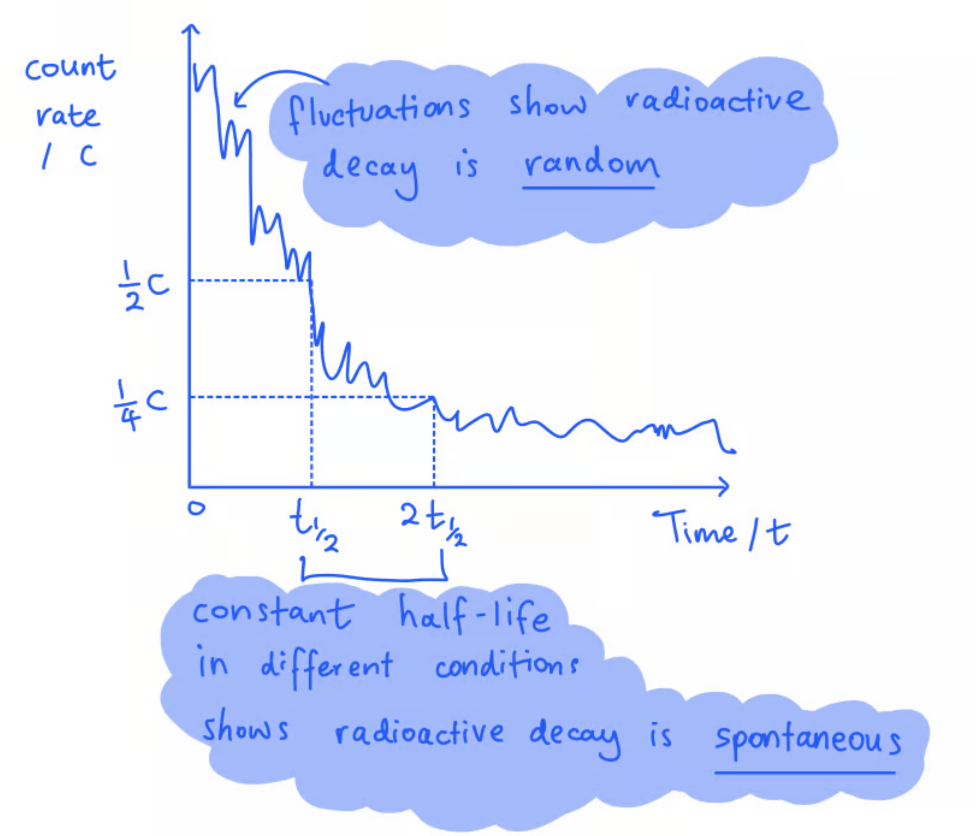
- Plotting a graph of count rate against time (using a data logger / computer software); [1 mark]
- To show the presence of fluctuations **OR** that the (decay) curve is not smooth; [1 mark]

OR max **one mark** for:

- (Random decay means) the time of decay (of a nucleus) cannot be predicted **OR** a nucleus has a constant probability (of decaying) in a given time; [1 mark]

[Total: 4 marks]

You could also draw a diagram or example decay curve to help illustrate your answer:



1b

(b) The correct equation for the decay is:

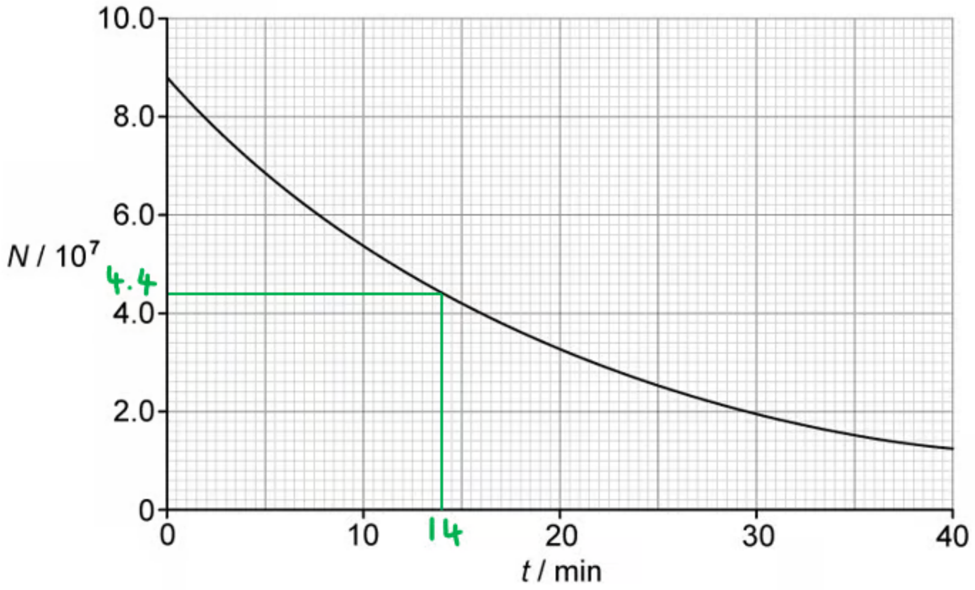
- ${}^{101}_{43}\text{Tc} \rightarrow {}^{101}_{44}\text{Ru} + {}^{0}_{-1}\beta$
- Ruthenium: A = 101, Z = 44; [1 mark]
- Beta-minus: A = 0, Z = –1; [1 mark]

[Total: 2 marks]

1c

(c)

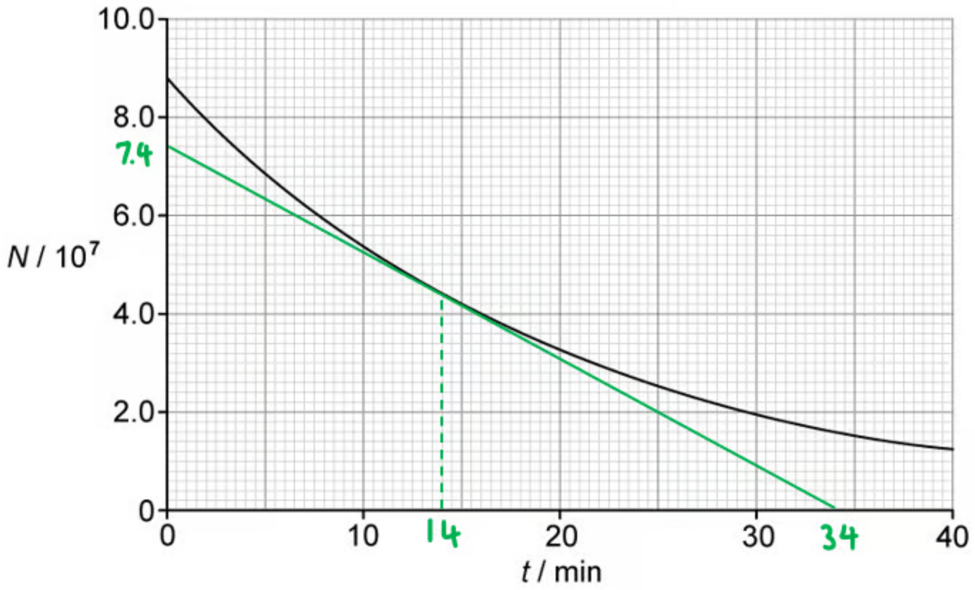
(i) Determine the activity of the sample of technetium-101 at 14.0 minutes:



EITHER

- (From graph) N at 14 minutes = 4.4×10^7 [1 mark]
- (From graph) t = 14 minutes is the half-life; [1 mark]
- Activity: $A = \lambda N \quad \Rightarrow \quad A = \dfrac{N \ln 2}{t_{1/2}}$ [1 mark]
- Activity: $A = \dfrac{(4.4 \times 10^7) \times \ln 2}{14 \times 60}$ = 3.6×10^4 Bq [1 mark]

OR

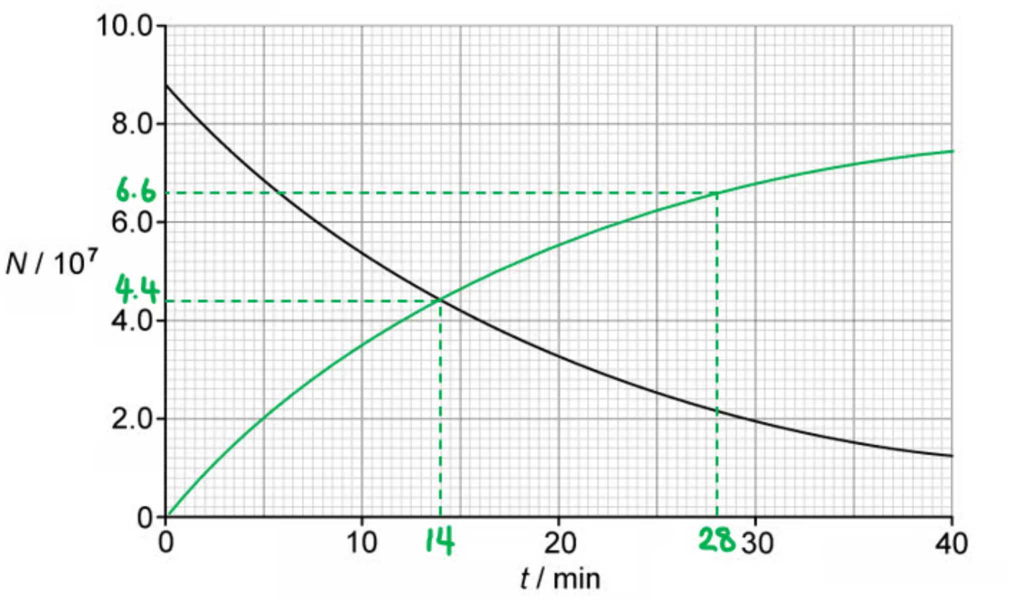


- Tangent drawn at t = 14 minutes; [1 mark]
- Activity = gradient of tangent; [1 mark]
- Activity = $\frac{(7.4 - 0) \times 10^7}{(34 - 0) \times 60}$ [1 mark]
- Activity = 3.6×10^4 Bq [1 mark]

(ii) Calculate the decay constant λ of technetium-101:

- $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{14 \times 60}$ [1 mark]
- $\lambda = 8.2 \times 10^{-4}\ s^{-1}$ [1 mark]

(iii) The variation with t of the number of ruthenium nuclei in the sample is as follows:



- Upwards curve of decreasing gradient starting from (0, 0); [1 mark]
- Passes through (14, 4.4) and (28, 6.6); [1 mark]

[Total: 8 marks]

1d

(d)

(i) Calculate the total amount of energy given to beta particles:

- (From graph) *N* at 10 min = 5.4×10^7 and *N* at 30 min = 2.0×10^7
- Total number of beta particles emitted = $(5.4 – 2.0) \times 10^7 = 3.4 \times 10^7$ [1 mark]
- Energy of a beta particle (in J) = 487 keV = $(487 \times 10^3) \times (1.6 \times 10^{-19})$ J
- Total energy released = $(487 \times 10^3) \times (1.6 \times 10^{-19}) \times (3.4 \times 10^7)$ [1 mark]
- Total energy released = 2.6×10^{-6} J [1 mark]

(ii) The total amount of energy released is greater because:

- Some Ru nuclei will have kinetic energy **OR** gamma photons are also emitted; [1 mark]

[Total: 4 marks]

2a

(a) The half-life of radium-228 is:

- The time taken for the number of atoms / nuclei / activity of radium-228; [1 mark]
- To be reduced to one-half (of its initial value); [1 mark]

OR

- In 5.75 years, the number of atoms / nuclei / activity of radium-228; [1 mark]
- Will be reduced to one-half of its current value; [1 mark]

[Total: 2 marks]

2b

(b) Calculate the decay constant of radium-228:

List the known quantities:

- Half-life of radium-228, $t_{1/2}$ = 5.75 years

Convert half-life into seconds:

- Half-life: $t_{1/2} = 5.75 \times (60 \times 60 \times 24 \times 365) = 1.81 \times 10^8$ s [1 mark]

- Energy of a beta particle (in J) = 487 keV = $(487 \times 10^{3}) \times (1.6 \times 10^{-19})$ J
- Total energy released = $(487 \times 10^{3}) \times (1.6 \times 10^{-19}) \times (3.4 \times 10^{7})$ [1 mark]
- Total energy released = 2.6×10^{-6} J [1 mark]

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[Total: 4 marks]

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- The time taken for the number of atoms / nuclei / activity of radium-228; [1 mark]
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OR

- In 5.75 years, the number of atoms / nuclei / activity of radium-228; [1 mark]
- Will be reduced to one-half of its current value; [1 mark]

[Total: 2 marks]

2b

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List the known quantities:

- Half-life of radium-228, $t_{1/2}$ = 5.75 years

Convert half-life into seconds:

- Half-life: $t_{1/2} = 5.75 \times (60 \times 60 \times 24 \times 365) = 1.81 \times 10^{8}$ s [1 mark]

Determine the decay constant:

- Decay constant: $\lambda = \frac{\ln 2}{t_{1/2}}$
- $\lambda = \frac{\ln 2}{1.81 \times 10^8}$ = 3.83×10^{-9} s^{-1} [1 mark]

[Total: 2 marks]

2c

(c)

List the known quantities:

- Decay constant of radium-228, $\lambda = 3.83 \times 10^{-9}$ s^{-1} (from **(b)**)
- Activity in 1.0 kg of water, A = 20 mBq = 0.020 Bq
- Avogadro's constant, $N_A = 6.02 \times 10^{23}$ mol^{-1}
- Mass of water in 1.0 mol, m = 18 g = 0.018 kg

(i) The number of radium-228 atoms in 1.0 kg of the contaminated water is:

- Activity: $A = \lambda N$
- Number of atoms in 1.0 kg of water: $N = \frac{A}{\lambda} = \frac{0.020}{3.83 \times 10^{-9}}$ [1 mark]
- Number of atoms in 1.0 kg of water: N = 5.22×10^6 [1 mark]

(ii) The ratio of molecules of water to the number of radium-228 atoms is:

- Number of water molecules in 1.0 kg = $\frac{N_A}{m} = \frac{6.02 \times 10^{23}}{0.018}$ = 3.34×10^{25} [1 mark]
- Ratio = $\frac{\text{number of molecules of water in 1.0 kg of water}}{\text{number of } ^{228}\text{Ra atoms in 1.0 kg of water}}$
- Ratio = $\frac{3.34 \times 10^{25}}{5.22 \times 10^6}$ = 6.4×10^{18} [1 mark]

[Total: 4 marks]

2d

(d) Calculate the time for the activity to be reduced to the safe limit:

List the known quantities:

- Target activity, $A = 18.5$ mBq kg^{-1}
- Initial activity, $A_0 = 20$ mBq kg^{-1}
- Decay constant of radium-228, $\lambda = 3.83 \times 10^{-9}$ s^{-1} (from **(b)**)

Use the exponential decay equation for activity to determine the time:

- Exponential decay: $A = A_0 e^{-\lambda t}$ [1 mark]

- $\frac{A}{A_0} = e^{-\lambda t} \quad \Rightarrow \quad \ln\left(\frac{A}{A_0}\right) = -\lambda t \quad \Rightarrow \quad t = -\frac{\ln\left(\frac{A}{A_0}\right)}{\lambda}$

- Time (in s): $t = -\frac{\ln\left(\frac{18.5}{20}\right)}{3.83 \times 10^{-9}}$ = 2.04×10^7 s [1 mark]

- Time (in days): $t = \frac{2.04 \times 10^7}{60 \times 60 \times 24}$ = 236 days [1 mark]

[Total: 3 marks]

3a

(a) Radioactive decay is defined as:

- The spontaneous emission of (ionising) radiation; [1 mark]
- From an unstable nucleus; [1 mark]

[Total: 2 marks]

3b

(b)

(i) The variation with t of the number N of nuclei of the isotope present in the sample is as follows:

- Curve drawn with decreasing negative gradient passing through (0, 1.00N_0); [1 mark]
- Curve passes through (T, 0.5N_0); [1 mark]
- Curve passes through (2T, 0.25N_0) **AND** (3T, 0.125N_0); [1 mark]

(ii) The name of the quantity represented by the gradient is:

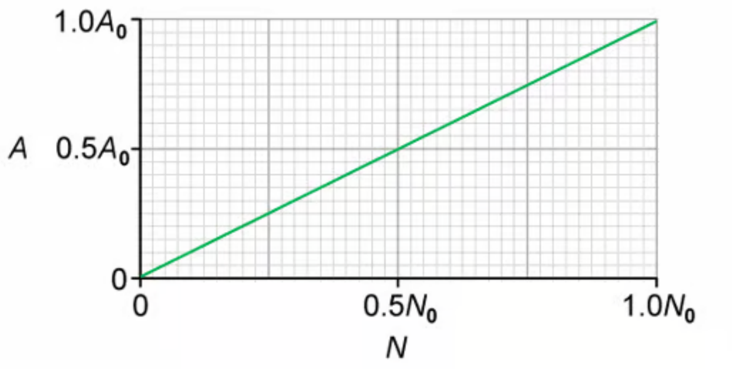
- The activity (of the sample); [1 mark]

[Total: 4 marks]

3c

(c)

(i) The variation with N of the activity A between N = 0 and N = N_0 is as follows:



- Line drawn through origin with positive gradient; [1 mark]
- Straight line drawn passing through (1.0N_0, 1.0A_0); [1 mark]

(ii) The name of the quantity represented by the gradient is:

- The decay constant; [1 mark]

[Total: 3 marks]

3d

(d) Calculate the fraction of undecayed nuclei N remaining relative to N_0 at time t = 1.25T.

List the known quantities:

- Half-life = T
- Time, t = 1.25T

Use the exponential decay equation:

- Decay constant: $\lambda = \frac{\ln 2}{T}$
- Exponential decay: $N = N_0 e^{-\lambda t} \quad \Rightarrow \quad \frac{N}{N_0} = e^{-\lambda t}$
- $\frac{N}{N_0} = e^{-\frac{\ln 2}{T} \times 1.25T} = e^{-1.25 \times \ln 2}$ [1 mark]

- $\frac{N}{N_0}$ = 0.42 [1 mark]

[Total: 2 marks]